2.2D Die last Integrated Substrate for High Performance Applications

Dyi Chung Hu
CEO Department
SiPlus Co.
Hsinchu, Taiwan, R.O.C.
hu@si2plus.com

Er Hao Chen
CEO Department
SiPlus Co.
Hsinchu, Taiwan, R.O.C.
eh.chen@si2plus.com

Jeffrey ChangBing Lee
Global Engineering Development Division
Integrated Service Technology Inc.
Hsinchu, Taiwan, R.O.C.
jeffrey_lee@istgroup.com

Chia Peng Sun
Product R&D Division
CoreTech System (Moldex3D)
Chupei, Taiwan, R.O.C.
victorsun@moldex3d.com

Chih Chung Hsu
R&D I Division
CoreTech System (Moldex3D)
Chupei, Taiwam, R.O.C.
jimhsu@moldex3d.com

2021 IEEE 71st Electronic Components and Technology Conference (ECTC) | 978-1-6654-4097-4/21/$31.00 ©2021 IEEE | DOI: 10.1109/ECTC32696.2021.00036

***Abstract*—In this work, a new integrated substrate structure "2.2D" is demonstrated. 2.2D substrate is a true die last solution. This solution used thin film RDL directly bonded to the substrate. This simplifies the 2.5D structure and further reduces the cost and improves the product cycle time. In this work, the 2.2D test vehicle is demonstrated with two metal layers of thin film that is bonded to a ceramic substrate. Serpentine lines run from thin film RDL to the substrate were evaluated. The process developed in this study demonstrates good registration between thin film RDL and substrate. The 2.2D TV demonstrates good warpage behavior up to the solder reflow temperature. The 2.2D structure shows good potential to be used for high performance computing substrate.**

Keywords— 2.0D, 2.1D, 2.2D, 2.3D, 2.5D, Silicon interposer, eHDF, CoWoS, RDL interposer, TSV-less, Solder-less, Integrated substrate, Heterogeneous Integration, Chiplet, High Performance computing.

I. Introduction

As society is embracing the age of AI, autonomous driving vehicles and smart-X (city, hospital, agriculture, etc.), the demand of high-performance computing power is ever increasing. Semiconductor is still following Moore's law by shrinking dimensions from 5nm, 3nm and 2nm and beyond. This drives up the need of interconnection density and of reducing the interconnection length between the connecting sites.

Traditional packaging cab be divided into different levels [1]. But due to the performance requirements, "level one" package (chip to substrate) is disappearing due to the benefit of reducing the connecting distance between chips by using bare dies. AMD's 2016 Fiji product uses bare dies for the interconnection between GPU and HBMs. This structure becomes the facto standard for the HPC packaging as shown in Fig. 1. In this case, 2.5D silicon interposers are used.

The term silicon interposer is interesting since silicon is only acting as the mechanical support of the fine line (down to submicron). Silicon manufacture has good infrastructure to meet the fine line requirements of the interposer. However, silicon is a semiconducting material that needs isolation of TSV (through silicon via). The TSV components can be modelled as resistance, capacitance and inductance [2]. Hence efforts have been made to cut the thickness of the silicon interposer or even to remove it completely. Various TSV-less solutions have been developed in the past. Examples are Amkor's SLIT [3], Samsung's RDL interposer [4] and TSMC's CoWoS-R [5].

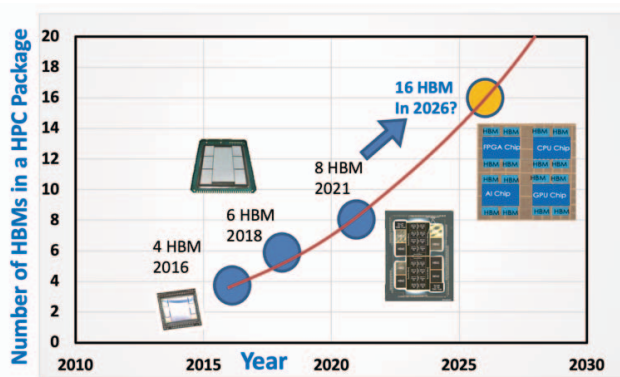


Fig. 1. The number of HBMs inside a substrate doubles every five years. Unofficial rule of HPC packaging.

Substrate with "coreless" structure has been in mass production over ten years due to the benefit of electrical performance of the system [6]. Solders are used for interconnection of silicon interposer and the underlying substrate. However, solder itself also can be modelled as resistance, capacitance and inductance [2]. Hence to remove the

2377-5726/21/$31.00 ©2021 IEEE
DOI 10.1109/ECTC32696.2021.00036

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

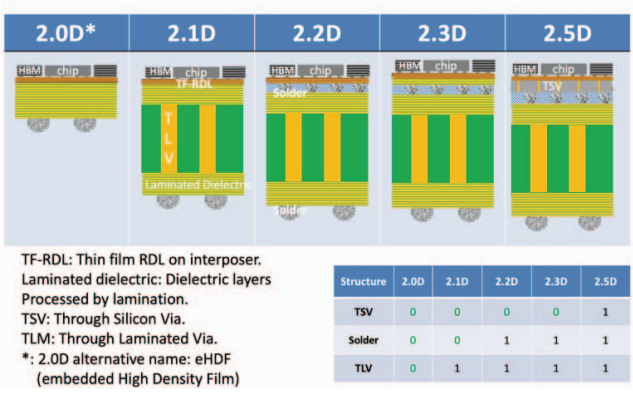
solders in the package would be beneficial to the electrical performance of the system. Solder-less solutions like FO-WLP, InFO-WLP and embedded die have been in production for many years.

There are various trade names used in the HPC packaging structure; namely 2.5D, SLIT, and CoWoS-S. It is desirable to classify the HPC packaging by their structure. Fig. 2 classifies HPC packaging by the components inside their structure, with increasing structure complexity from 2.0D constitute to to 2.5D. For the 2.5D structure, silicon interposer with TSVs are connected to the underlayer cored substrate with solders. The cored substrate has TLVs (Through Laminated Vias). On the other hand, the 2.0D packaging is a TSV-less, TLV-less and solder-less structure that can take the full benefit of the system electrical performance. The 2.0D TV's manufacturability and reliability have been verified by SiPlus and Nanya PCB Co. [7].

The advanced 2.0D structure that combines BEOL, thin film RDL (TF-RDL) and laminated dielectric layers (ABF, PP) is constituted to one "integrated substrate". This integrated substrate can be used for the die last chiplets and dielets applications in Fig. 3 [8].

RDL interposer (Amkor/Samsung) and CoWoS-R (tsmc) are similar to the 2.2D structure after chip assembly. However, those are "die middle" and not a true die last solution. Since dies are bonded to RDL and then mold, de-bond before joining to the substrate. It is desirable to have a substrate that is a true 2.2D die last solution.

In this paper, an innovation 2.2D die last solution to enable TF-RDL directly on substrate is proposed. Test vehicles have been built to test its feasibility and manufacturability.



Structure	2.0D	2.1D	2.2D	2.3D	2.5D
TSV	0	0	0	0	1
Solder	0	0	1	1	1
TLV	0	1	1	1	1

Fig. 2. Classification of HPC according to their packaging structure. Note that from 2.0D, 2.1D to 2.5D with increasing packaging structure complexity.

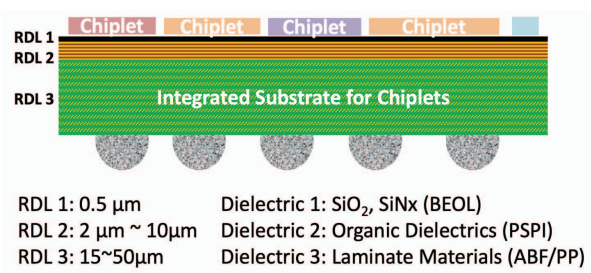


Fig. 3. Advanced 2.0D structure that combines BEOL, TF-RDL and laminated dielectric into one high performance 2.0D substrate that can be used for dielets and chiplets applications.

II. 2.2D Test Vehicle Design

To test the feasibility of the 2.2D structure, three important items need to be verified. (1) The dimensional stability of the TF- RDL needs to be maintained, otherwise there will be mis-registration between thin film RDL and substrate. (2) The robustness of the joints between TF-RDL and substrate need pass the substrate reliability test. (3) It is better to take the advantages of the existing infrastructure to realize the low cost manufacture.

There are several joint structures and surface metallization between TF-RDL and substrate are possible. For example, pillar connects to pad with solder on top, pillar to pillar connection with solder, or Cu to Cu direct bonding etc. There are also several surface metallization options, such as Cu to Cu, Au to Au, Au to solder, Au to Sn, etc. Furthermore, there are options for substrate material selections, for example, organic, ceramics and glass substrate are under consideration.

In this study, copper pad with solder plating is the structure at the TF-RDL side and copper pillar with Au surface finish on ceramic is at the substrate side.

To check the connectivity of TF-RDL directly to the substrate, 3x3mm serpentine loops on TF-RDL and on ceramic substrate are designed. Fig. 4 shows the CAD design pattern of the TF-RDL. The thin film pads with diameter of 400µm and pitch 500µm are designed. 5µm Asahi Kasei low temperature curing Polyimide are used as thin film dielectric layer. 3µm Copper metallization are used for conduction layer with Ti/Cu seed layer underneath. There is a lead free solder plating on top of each TF-RDL pads.

AlN ceramic substrate with 1mm thickness was chosen as a substrate. To construct the vertical serpentine line, pillars on the AlN substrate are designed to be connected in pairs (Fig. 5).

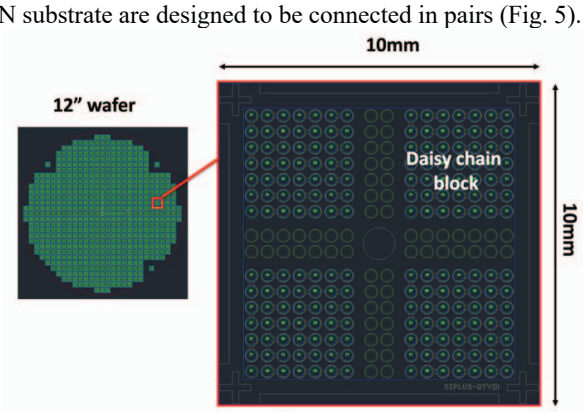


Fig. 4. The thin film CAD test pattern designs.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

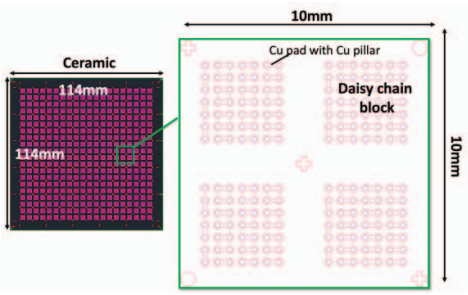


Fig. 5. CAD pattern designs on the ceramic substrate.

Thermal compression bonding is generally used for the interconnection of large die to a substrate. However, conventional TCB tools are limited to reticle size (33x26mm) die bonding. The goal of this paper is to achieve building a fine line 2.2D substrate with a substrate size that is bigger than the reticle size. For easy manufacture of the 2.2D substrate, adapting thermal reflow in conventional solder reflow oven is desirable. However, a good planarity is required for the interconnection between thin film and substrate during the 260°C reflow temperature. Since TF-RDL is on a glass substrate that is quite flat, the pillars on the ceramic substrate are keeping flat by planarization after plating the copper pillars.

There are serval options to protect the joint between TF-RDL and pillars on ceramic substrate after the joining process. Namely the capillary underfill, film typed underfill, molded underfill etc. In this paper, capillary underfill was chosen.

III. 2.2D Integrated Substrate TV Manufacture Process and Results

The 2.2D integrated substrate TV manufacture process flow of this work is shown in Fig. 6. Fig. 7(a) shows the copper pillars manufactured on a AlN substrate. Where the upper part of the pillar are electroplated with 5µm Ni and 0.3µm Au after the copper pillars were planarized.

Two layers of Polyimide (PI) dielectric and two layers of Cu metal were built on a 12" glass wafer. Each 10x10mm pattern size has four 3x3mm daisy chain blocks and can be treated as one basic unit. The basic units were populated all over on the 12" wafer. Both patterns (Figs. 7(a) and 7(b)) are designed with corresponding locations to form the continuous daisy chains after surface mount process. For easy visual illustration of the interconnection between TF-RDL and substrate copper pillar, Fig. 7(c) is a TF-RDL pattern overlay on top of the ceramic substrate pattern. There is lead free solder on top of each TF-RDL pad. The Cu pillars were built with height of 80µm on top of the ceramic substrate. The height variations of solder and Cu pillar are controlled to be within +/- 5µm and +/-10µm, respectively by the manufacture. The nominal dimensions of the critical patterns are listed in TABLE I.

A. 2.2D substrate assembly Process

One of the innovative design of the 2.2D TV pattern in this study is that it can accommodate 2.2D substrates with incremental size of 10mm. Fig. 8 demonstrates that the integration of substrate with three different sizes can be placed at will on the surface of a 12" TF-RDL glass wafers. In this paper, the joining TF-RDL to the substrate with sizes of 20x20mm, 40x40mm and 60x60mm have been evaluated. One 60x60mm sample is used to check the feasibility of surface mounting process to a large substrate. The 40x40mm samples are used to get the warpage information of 2.2D integrated substrate. And the 20x20mm samples are prepared for the reliability qualification test.

The ceramic substrates were placed on a fluxed 12" glass wafer by a pick and place equipment with placement accuracy of +/-25µm. Later, the assembly went through a conventional furnace with peak temperature of 260°C for solder reflow.

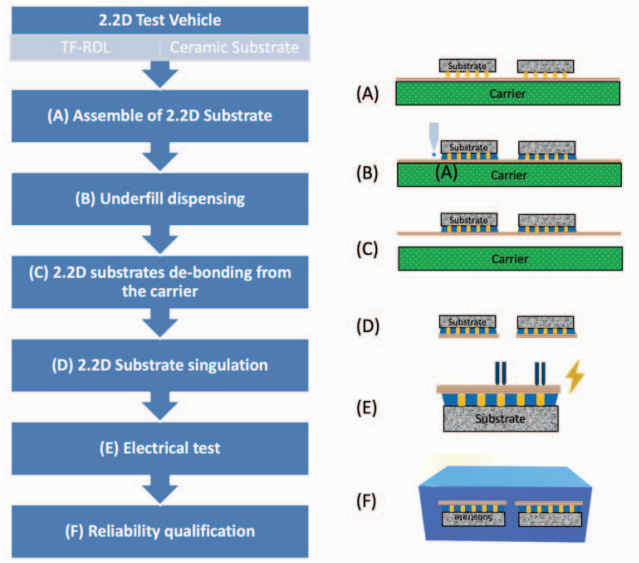


Fig. 6. Process flow of 2.2D integrated substrate TV.

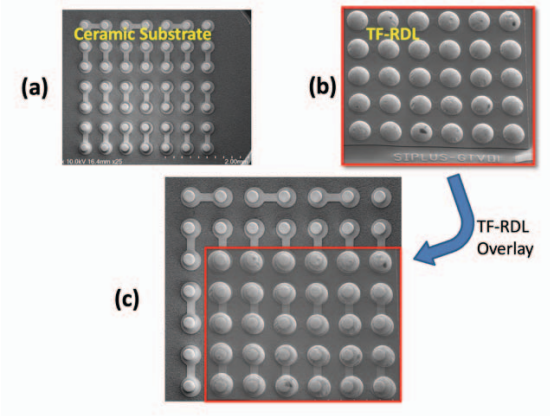


Fig. 7. SEM pictures of (a) Test patterns on ceramic substrate. (b) Test patterns on thin film. (c) is a TF-RDL pattern overlays on top of the ceramic substrate.

TABLE I. The Nominal Dimensions of the Critical Patterns

Pattern	Cu pad THK (µm)	Cu pad diameter (µm)	Cu pad pitch (µm)	Solder height (µm)	Solder diameter (µm)	Cu pillar height (µm)	Cu pillar diameter (µm)
Thin flm	3	400	500	50	350	NA	NA
Ceramic substrate	15	350	500	NA	NA	80	200

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

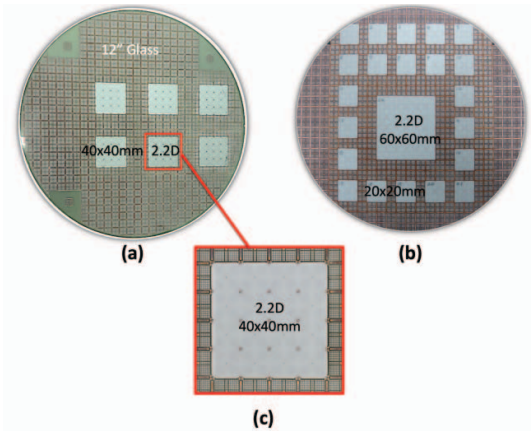


Fig. 8. Images for (a) Six 40x40mm ceramic substrates are bonded to a 12" TF-RDL glass wafer (b) One 60x60mm and 21 20x20mm ceramic substrates are mounted on a 12" pattern glass wafer. (c) A magnified view of one 40x40 mm 2.2D substrates.

The daisy chain test patterns are used to check the connectivity of the TF-RDL to the ceramic substrate. The schematic structures of thin film and ceramic substrate after surface mount process are shown in Fig. 9. This 3D metal structure on the ceramic substrate is produced by etching off the two layer polyimide dielectric. The green line indicates the electrical path from the TF-RDL pad through the copper pillar on the ceramic and move to the adjacent copper pillar connected to the top TF-RDL pad. There are total of 7x7 TF-RDL connection pads with matching copper pillars on the ceramic substrate.

B. Underfill dispensing

After the 2.2D substrate assembly process, underfill is dispensed to protect and enhance the bonding strength between TF-RDL to the ceramic substrate. Since the 40x40 mm 2.2D substrate size is bigger than the ordinary chips, a "L" shape dispensing pattern is used. The calculated underfill volume is shown in TABLE II. Use 40x40mm sample size as an example, each ceramic substrate needs to do 14 times of line dispensing to finish the underfill process.

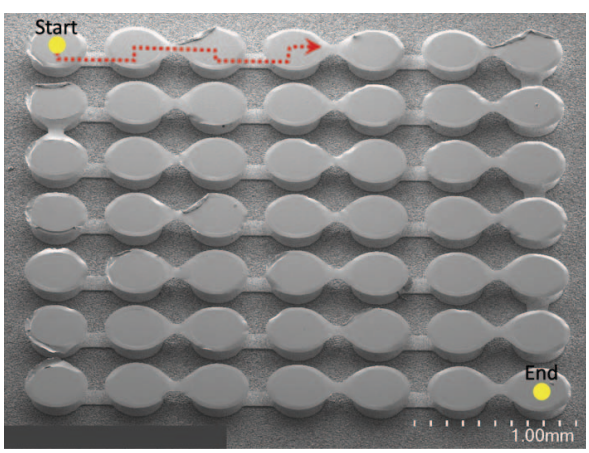


Fig. 9. The schematic structures of 2.2D TV integrated substrate. The thin film dielectric layers are removed by etching for easy visualization. Note that since there is no polyimide dielectric support, several pads edge of TF-RDL are deformed under stress.

TABLE II. THE CALCULATED UNDERFILL VOLUME FOR 40X40MM 2.2D SUBSTRATE.

Item	Volume (cm^3)
V_total inner space	0.2608
V_total ceramic substrate	0.0137
V_total thin film	0.0215
V_remaining space	0.2256

The underfill material A is the one that used for packages on PCB. It has a low viscosity for easy underfilling. After filling, the assembly was cured in oven at 150°C for 15 mins. The images after underfilling are shown in Fig. 10.

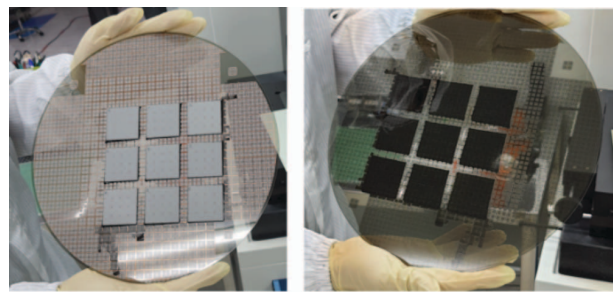


Fig. 10. Images on the left: Underfilled 2.2D substrate, from substrate side view. Images on the right: Underfilled 2.2D substrate, from thin film side view.

The amount of underfill quantities that are needed in dispensing process can be calculated by subtracting of the volumes occupied by the joint structures, i.e., the heights of thin film Cu pad (3μm), ceramic substrate Cu pad (15μm), Cu pillar (80μm), Nickel (5μm), PI (10μm) and solder (50μm). The calculated total inner space is 0.2608cm^3, the occupied volumes from thin film and ceramic substrate patterns are 0.0215cm^3 and 0.0137cm^3, respectively. Therefore, the remaining space for underfill is 0.2256cm^3 which is quite close to the real dispensed underfill quantity 0.218cm^3, with only ~4% difference.

To further implement the underfill process for the 2.2D substrate; a good simulation tool is needed to optimize the copper pillar height, underfill gap and underfill material selection. CAE tool from Moldex3D [9] is used for the simulation the underfill process. Before the simulation, measurement of underfill material properties at the process temperature are needed. Figs. 11(a) and 11(b) show the measurement of underfill material A's viscosity and curing kinetic vs. temperature respectively. Figs. 11(c) and 11(d) show the physical model of this study. To simulate the real underfill process; three I-type passes are set by 5mg per pass, the drop size is 0.9 mm^3 with a pass duration of 20 sec. The simulation of flow propagation and fillet height from each pass by underfill material A are shown in Fig. 12. The flow propagation shows the melt front near edge moves faster than the middle zone of the gap. It shows that the surface-tension effect of the solders is too small to lead the flow for current 2.2D TV. The simulation model will be continuing fine-tuned in the future.

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

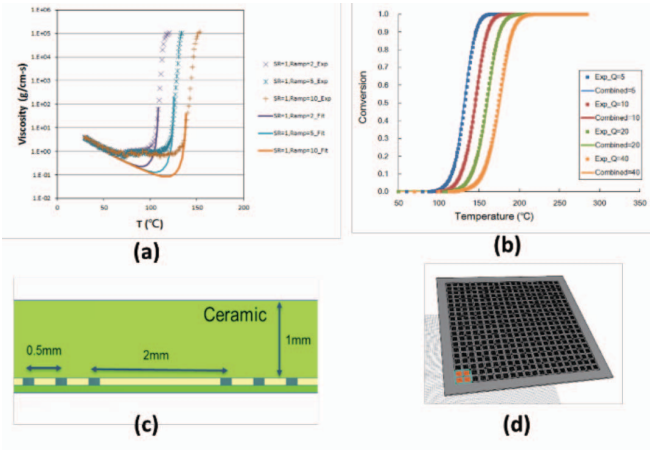


Fig. 11. (a) viscosity vs. temperature @ shearing rate =1.0 (b) Curing kinetics. (c) and (d) Geometry of the 2.2D TV case.

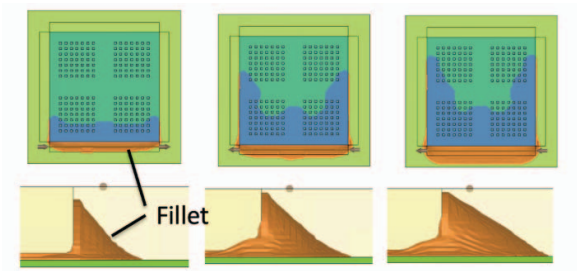


Fig. 12. Underfill simulation results of flow propagation (top) and fillet height (bottom).

C. 2.2D Substrate De-bonding from the 12" glass carrier

After underfill dispensing process, laser de-bonding is used to remove the 12" glass carrier wafer. Fig. 13 shows TF-RDL with ceramic substrates is transferred to a blue tape.

D. 2.2D Substrate Singulation

All the ceramic substrates mounted on TF-RDL are singulated into 2.2D integrated substrates by laser cutting. Fig. 14 shows the front and back side of one singulated 40x40mm 2.2D integrated substrate.

The warpage behavior of 2.2D integrated substrates with size 40x40mm were evaluated by using Shadow Moire test. Three test samples (#2, #3 and #6) and one temperature monitoring sample (#9) are placed in the platform (as shown in Fig. 15) to undergo the temperature cycling from room temperature to 260°C and then again cool down to the room temperature. The 3D warpage maps by using sample #2 as an example is shown in Fig. 16. There are extra peaks occurred in the corner site (marked in red circle). This is caused by the bubbles entrapped inside the underfill material. The bubbles are easy revealed under the microscope since the PI is translucent. The cause of the entrapped bubble is during the underfill dispensing process. The volume of entrapped bubbles are expending with the temperature raising during the Shadow Moire test. The top right corner that bubbles induced protrusion is the site that opposite to the L shape dispensing (as shown in Fig. 15(b)). In that place, the bubble escaping routes were blocked during the 13th and 14th underfill dispensing process.

The warpage variation along the diagonal line is measured from the top left to lower right corners. The variation is less than +/- 20μm as show by the rectangular green area in Fig. 16. That means except the top right corner that the extrusions caused by the entrapped bubbles, other areas show good planarity through the temperature cycling. The +/- 20μm warpage is the measurement across 40mm range, that means +/- 10μm across 20mm range, that is good for fine pitch die attachment.

The issue of entrapped bubbles can be improved by (1) change the dispensing sequence i.e., remove dispersion path 13 and 14. (2) Dispensing under reduced atmosphere or vacuum. (3) Use pressure chamber to remove the bubbles before the curing the underfill materials.

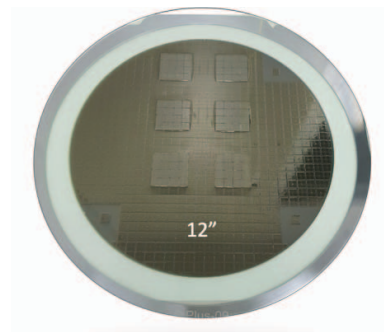


Fig. 13. TF-RDL with ceramic substrates is transferred to a blue tape.

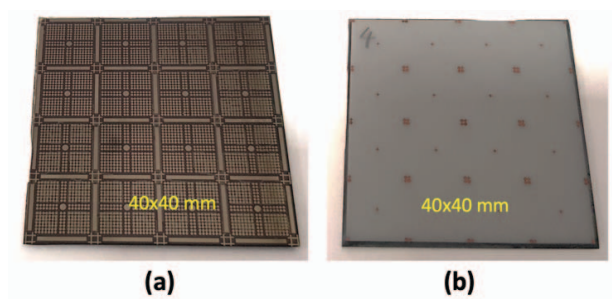
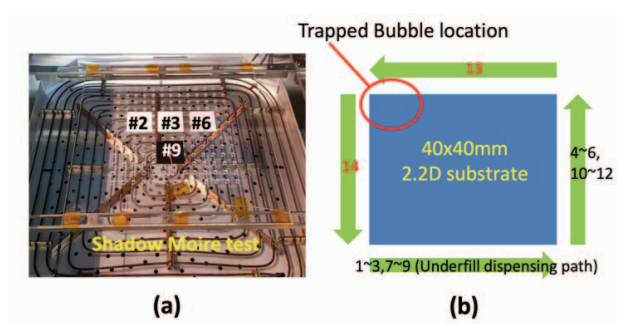


Fig. 14. 40x40mm 2.2D integrated substrate (a) front-side (b) back-side.

The microstructure of 2.2D substrate can be revealed in detail by using SEM and 3D X-ray shows in Figs. 17 and 18. Fig. 17 shows the 3D X-ray image; top view 17(a), cross section views along the green line 17(b), along the red line 17(c). The formation of solid solder joins are clear. There are several small voids inside the solder join that can be revealed as black dots on a white background of solder under X-ray examination.



Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

Fig. 15. Shadow Moire test (a) samples placed inside the test chamber. (b) Locations of entrapped of bubble.

Fig. 18(a) is a SEM cross section view of a 20x20mm 2.2D substrate. TF-RDL is not visible under this magnification. The magnified portion of two connecting pillars is shown in Fig. 18(b). The pitch between two copper pillars is measured to be 496µm that is very close to the design value of 500µm. The formation of solid solder joins are clear. There are several small voids inside the solder join that can be revealed as black dots on white background of leadfree solder under the X-ray examination.

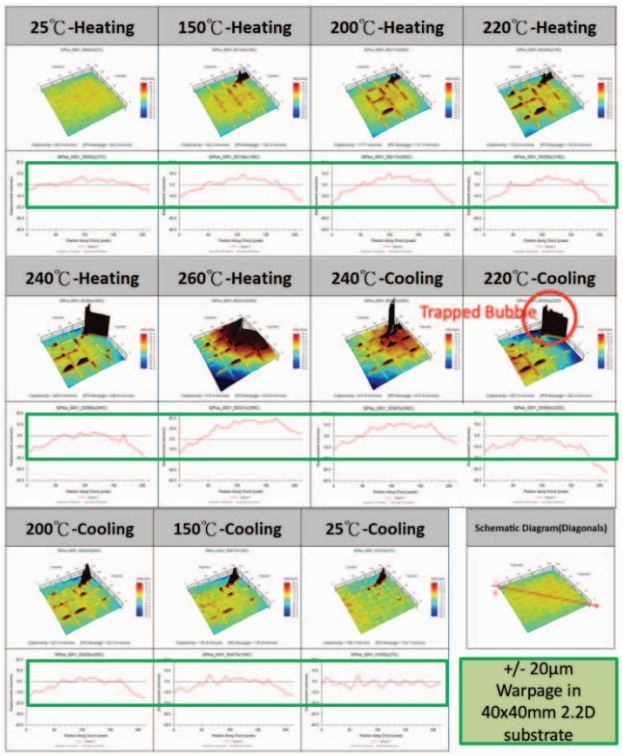


Fig. 16. Shadow Moire 3D warpage maps for sample #2. AXP 2.0 is used for Shadow Moire test.

There is delamination between underfill and solder in Fig. 18(b). This can be caused by the high temperature (>300°C) CP (Cross-section Polisher) sample preparation process for SEM. The underfill material "A" that we used in this study has a low filler content. The filler particles are settled to the TF-RDL side. There are several small voids inside the solder joins. These voids can be reduced if the soldering process is under reduced atmosphere or vacuum.

E. Electrical measurement

To perform the subsequent E-test on the 2.2D integrated substrate, the PI dielectric layer on top of the TF-RDL pads need to be removed. Hence, the laser drilling process is performed. Two drilled holes to reach TF-RDL pads in each 3x3mm daisy chain block are opened by laser.

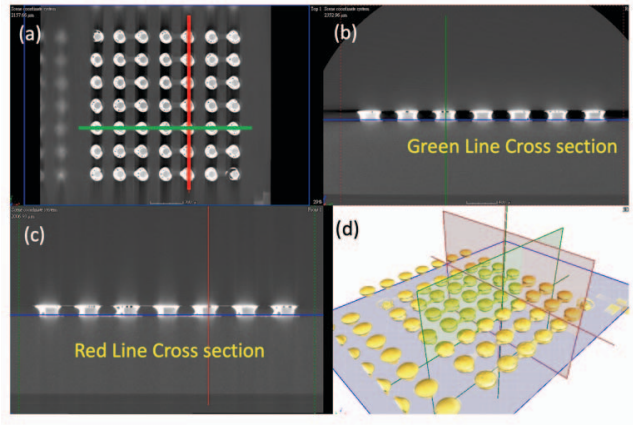


Fig. 17. 3D X-ray image of the 2.2D substrate. (a) Top view (b) Cross section view of the joints along the green line. (c) Cross section view of the joints along the red line. (d) Another angle of green and red line cross section of the 2.2D joints. Xradia 620 is used for the 3D X-ray image.

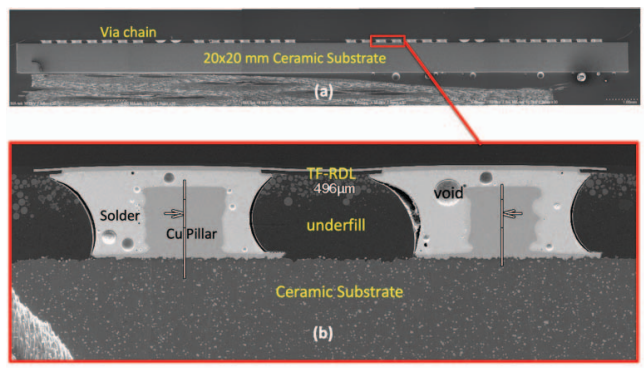


Fig. 18. (a) Cross section view of a 20x20mm 2.2D substrate. (b) SEM picture shows two copper pillars that are connected by thin film RDL on top. Several small voids in the solder joints can be seen.

Each 10x10mm pattern unit has 4 daisy chain blocks. For 20x20mm and 40x40mm samples, daisy chain blocks located in the two crossed diagonals are measured. For 60x60mm sample, except the blocks in the two crossed diagonals, 4-side peripheral ones are also tested. The test results are plotted in accumulation with rather tight distributions as shown in Fig. 20. The averaged resistances for 20x20mm, 40x40mm and 60x60mm samples are 0.366Ω, 0.336Ω and 0.347Ω, with standard deviations 0.035Ω, 0.027Ω and 0.021Ω, respectively. As compared with the theoretical value 0.356Ω (calculated in TABLE III), the difference percentages are only 2.8%, 5.8% and 2.6%, respectively. The theoretical resistance value is calculated by using the law of resistance $R = \rho\ (l/A)$ based on the dimensions of dog-bone shapes and the vertical interconnects. From the calculation data, it shows the daisy chain resistance mainly comes from the thin film patterns since the Cu layer thickness is only 3µm, that is one fifth of that in ceramic substrate copper trace thickness. Over 80% resistance is contributed from thin film RDL side and nearly 16% from the ceramic substrate side. Only 1~2% comes from the vertical interconnects (i.e., Cu pillars and solders). The voids inside the solder seem to have a very minor effect to the overall resistance in this study.

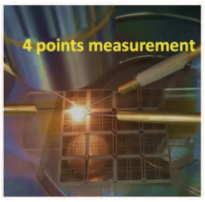
Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.

F. Reliability test

20x20mm 2.2D substrate TVs are under MSL4/245°C X3 test. The averaged via chain resistivity change is less than 3% of the value prior to the MSL4 test.

However, the CSAM data show several delamination and voids of underfill occur in the samples after MSL4 test. Hence more efforts to the selection of underfill materials and surface treatment prior to the underfill process are needed to be fine-tuned in the future.

TABLE III. THEORETICAL CALCULATION OF THE RESISTANCE DATA



Item	Resistance (Ω), R = ρ(l/A)
Single dog-bone pattern, R_TF	1.27E-02
Single dog-bone pattern, R_Ceramic	2.55E-03
Single vertical interconnect, R_solder	5.98E-05
Single vertical interconnect, R_Cu pillar	4.28E-05
R_total	3.56E-01

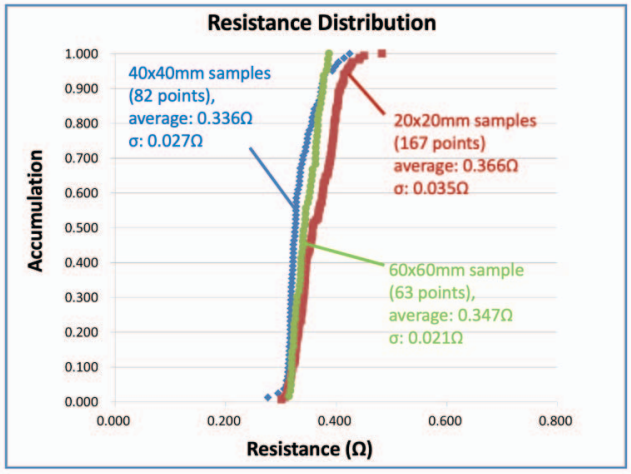


Fig. 19. E-test results for 60x60mm, 40x40mm and 20x20mm 2.2D integrated substrate samples.

IV. CONCLUSION

The structure and manufacturability of 2.2D substrate have been demonstrated in this paper. We used an creative process to avoid thin film deformation and make 2.2D integrated substrate a reality. 2.2D substrate technology is an important technology that bridges the RDL first technology with the conventional substate. The 2.2D substrate technology has potential to extend the current organic laminated substrate line width down to 2µm by including the thin film RDL technology.

2.2D test vehicles of sizes 20x20, 40x40 and 60x60mm are built with creative TF-RDL pattern design. The connectivity of TF-RDL to ceramic substrate have been verified by 4 point measurements of daisy chain test patterns. A large fine line substrate size of 60x60mm TV also shows good connectivity. The tight and uniform distributions of the E-test results for different sizes of 2.2D integrated substrates show the good controllability in assembly of the thin film and ceramic substrate. The warpage of the 2.2D integrated substrate is within +/-20µm in 40mm range from room temperature up to 260°C. Hence the fine pitched dies have potential to be integrated on top of the 2.2D substrate by using ordinary solder reflow process. Another benefit of the 2.2D integrated substrate is that one can select know-good TF-RDLs and know-good substrates before assembly. By doing so, one can have both good yield and reducing the time to the market. Since TF-RDL and substrate can be built in parallel.

There are more developing work needs to be fine tunned to enable the 2.2D integrated substrate for mass production. Nevertheless, this work demonstrates 2.2D integrated substrate has potential to be used for the large substrate with fine lines applications.

ACKNOWLEDGMENT

The authors would like to thank "iST RA lab for supporting the surface mounting process, underfill dispensing, MSL precondition and thermal warpage measurement by Shadow Morie, and FA lab for supporting 2D/3D X-ray, CSAM, SAT and CP/SEM.

REFERENCES

[1] Donald Serephim, Ronald C. Lasky, Che-Yu Li, "Principle of Electronic Packaging", McGraw-Hill College, 1989.

[2] Madhavan Swaminathan and Ki Ji Han, "Design and modeling for 3D ICs and Interposers", World Scientific Publishing Co. Pte. Ltd. 2014.

[3] Young Rae Kim et all, "SLIM", High Density Wafer Level Fan-out Package Development with Submicron RDL", 2017 IEEE, 67th ECTC Conference.

[4] K.L. Suk, S.H. Lee, J.Y. Kim, S.W. Lee, H.J. Kim, S. C. Lee and et al., "Low Cost Si-Less RDL Interposer Package for High Performance Computing Applications," 2018 IEEE, 68th ECTC Conference.

[5] Yi-Hang Lin, M.C.Yew, M.S. Liu , S.M. Chen, T.M. Lai, and et al., "Multilayer RDL Interposer for Heterogeneous Device and Module Integration" 2019 IEEE, 69th ECTC Conference.

[6] https://www.shinko.co.jp/english/product/ic-package/buildup/dll3.php

[7] Dyi-Chung Hu and James Ho, "Methods to Reduce the Hierarchy of Interconnections in Electronic System", Proceedings of IMAPS 2020.

[8] Dyi-Chung Hu, "An Innovative System Integration Interconnection Technology Beyond BEOL" IEEE International Interconnection Conference, Santa Clara, USA, 2018

[9] W. H. Yang and R. Y. Chang, "Numerical Simulation of Mold Fill in Injection Molding Using A Three-Dimensional Finite Volume Approach", International Journal for Numerical Methods in Fluids, vol. 37, p. 125,

Authorized licensed use limited to: JILIN UNIVERSITY. Downloaded on December 10,2024 at 14:36:39 UTC from IEEE Xplore. Restrictions apply.